

1) Newtons Cooling Law

$$\frac{dT}{dt} = K(T - T_e)$$

a differential equation is an equation containing derivatives of 1 dep. var wrt 1 indep. var

the order of a DE is the highest derivative appearing

$$\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} - xtyz$$

2nd order; y & z = dep & x & t are indep.

initial value problem

$$\begin{cases} dy/dt \\ y(t_0) = y_0 \end{cases}$$

$$y(t) = \sin t + y_0 - \sin(t_0)$$

$$r^2 e^{rt} + q e^{rt} = 0$$

$$y(t) = c_1 \cos(3t) + c_2 \sin(3t)$$

$$e^{rt}(r^2 + q) = 0$$

$$r = \pm 3i$$

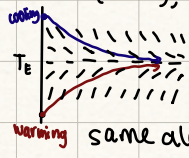
When solving a DE is difficult, the DE itself can be used to tell us about sol. behavior

slope fields (direction fields)

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example of slope fields

$$\frac{dT}{dt} = K(T - T_e), K < 0$$

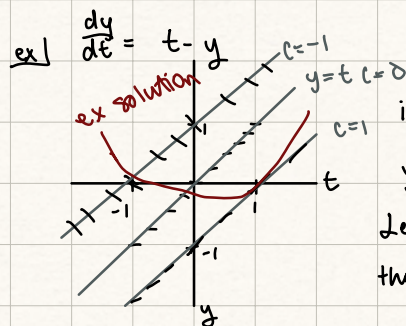
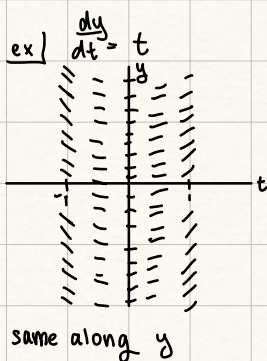


note: in this case, the DE is autonomous

↳ slope fields not dependent on t

same along t

* jupyter notebook *



isoclines: $\frac{dy}{dt} = C$

$$y = t - C$$

Lets say $C = 0$ } this specific case
then $y = t$ } is a nullcline

numerical methods

→ approx solutions to 1st order DE

→ derivation of method, method, error of approx & failures

↳ largely by Taylor Expansion

Euler's Method → starting point →

goal: (1) Approx solutions to 1st order DE of the form $\frac{dy}{dt} = f(t, y)$ on the grid: $t_0, t_1, t_2, \dots, t_n$ subject to the initial condition $y(t_0) = y_0$.

(2) local + global error: $\text{Global Error} = \left(\frac{h^2}{2} y''(c)\right) n = \frac{h^2}{2} \left(\frac{t_n - t_0}{h}\right) y''(c) = O(h)$
 $n = \frac{t_n - t_0}{h}$

NOTATION let w_i be the approx to the exact solution y_i . here $w_i = w(t_i) \approx y_i = y(t_i)$

How do we approx @ t_i , $i=1, \dots, n$

↳ Taylor Expand around $t_0 \Rightarrow y(t_0+h) = y(t_0) + y'(t_0)h + \frac{h^2}{2} y''(t_0) + \frac{h^3}{3!} y'''(t_0) + O(h^4)$
 $O(h^3)$ ignore...
 $O(h^4)$ local error

$$y(t_0+h) = y(t_0) + hf(t_0, y(t_0)) + O(h^2)$$

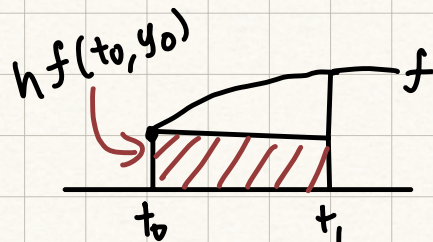
Euler's method: $w_{i+1} = w_i + hf(t_i, y_i)$

another derivation...

$$\frac{dy}{dt} = f(t, y) \quad \int_{t_0}^{t_1} \frac{dy}{dt} dt = \int_{t_0}^{t_1} f(t, y(t)) dt$$

$$y(t_1) - y(t_0) = \int_{t_0}^{t_1} f(t, y(t)) dt$$

$$= y(t_0) + \int_{t_0}^{t_1} f(t, y(t)) dt$$



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generally if a method has a local error $O(h^p)$, then global error is $O(h^{p-1})$

code validation:

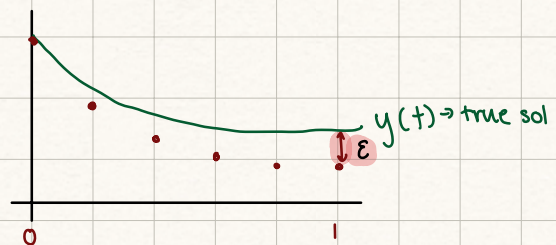
$n = 10, 100, 1000, 10000$

global error: $O(h)$ or $O(h^{-1}) \rightarrow E = cn^{-1}$

$$\ln(E) = \ln(cn^{-1}) = \ln(c) + \ln(n^{-1}) = \ln(c) - \ln(n)$$

$$\ln(E) = E = C - N \quad \text{where } \ln(c) = C \text{ \& } \ln(n) = N$$

slope of error is $p-1 \Rightarrow$ behaves like a line with slope -1



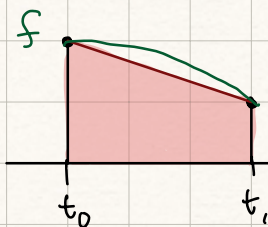
Consider: $\frac{dy}{dt} = f(t, y)$, $y(t_0) = y_0$

$$\int_{t_0}^t \frac{dy}{dt} dt = \int_{t_0}^t f(t, y(t)) dt$$

$$y(t_1) - y(t_0) = \int_{t_0}^{t_1} f(t, y(t)) dt$$

numerical integration

approx $f(t, y(t))$ with linear function



$$= \frac{h}{2} (f(t_0, y_0) + f(t_1, y_1)) + \frac{h^3}{12} f'''(c) \quad (O(h^3))$$

Trapezoid Method $\Rightarrow$ the implicit trapezoid method

$$w_{i+1} = w_i + \frac{h}{2} (f(t_i, y_i) + f(t_{i+1}, w_{i+1})) + O(h^3)$$

We must go from implicit $\rightarrow$ explicit for w_{i+1}

$\hookrightarrow$ We can use Euler's method to approximate $w_{i+1} = w_i + h f(t_i, w_i)$

the explicit trapezoid method:

$$w_{i+1} = w_i + \frac{h}{2} (f(t_i, w_i) + \overbrace{f(t_{i+1}, w_i + h f(t_i, w_i))}^{\text{euler's method}}) + \epsilon$$

need to compute local error for implicit trapezoid

let $w_i = y_i$ and compute $y_{i+1} = w_{i+1}$

$\rightarrow$ expression for $y_{i+1} = y(t_i + h)$

$$y(t_i + h) = y_i + h y_i' + \frac{h^2}{2} y_i'' + \frac{h^3}{6} y_i''' + O(h^4) \quad \text{doesn't cancel}$$

$$y_{i+1} - w_{i+1} = \overbrace{y_i + h y_i'}^{\text{cancel}} + \frac{h^2}{2} y_i'' + \frac{h^3}{6} y_i''' + O(h^4) -$$

$$(y_i + \frac{h}{2} (y_i' + y_{i+1}')) \quad y_{i+1}' = y_i' + h y_i'' + \frac{h^2}{2} y_i''' + O(h^3)$$

$\therefore$ this error is $O(h^3)$